Solving For minimum $\varnothing$:

Knowns:

- Load capacity (F_{TEN}) = 2 kN (Given)
- FoS = 4 (Given)
- Crosshead speed: 1-50 mm/min (Given range)
- Leadscrew height/length (L) = 18 in minimum → (use as min length (L))
- 2 Leadscrews → F_{lead} = 1 kN
- Assumed Both Ends of Tensile Tester are Fixed

Unknowns:

- σ_{yield} (Based on Material Selection)
- σ_{allow}
- $\varnothing_{min}$: Area (A)

Material Selection:

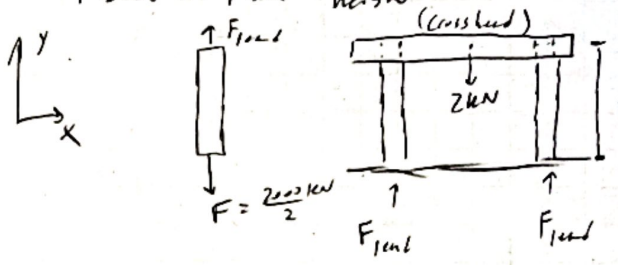
Medium carbon Steel

- σ_{yield} = 684 (MatWeb Average value) MPa

Determining σ_{allow}:

$$\sigma_{allow} = \frac{\sigma_{yield}}{FoS} = \frac{684\,MPa}{4} = \underline{171\,MPa}$$

FBD: at Max height



$\Sigma F_y = 0 = F_{lead} + F_{lead} - 2kN$

$F_{lead} = 1kN$

Tension in crosshead → Compression in leadscrew

Solving For $\varnothing_{min}$:

$$\sigma_{allow} = \frac{\sigma_{yield}}{FoS} = \frac{F_{lead}}{A} \quad (eq\,1)$$

$$A = \frac{\pi}{4}\varnothing_{min}^2$$

$$\frac{\sigma_{yield}}{FoS} = \frac{F_{lead}}{\frac{\pi}{4}(\varnothing_{min})^2}$$

(eq 1) $$\varnothing_{min} = \sqrt{\frac{FoS\,(F_{screw}) \cdot 4}{\pi \cdot \sigma_{yield}}}$$

$$\varnothing_{min} = \sqrt{\frac{4(1000N)\cdot 4}{\pi(684\cdot 10^6 Pa)}} = 0.00273m$$

$$\varnothing_{min} = 0.107\,in$$

Solving For Range of Rotational Speed (W)

Knowns:

- Cross Feed Speed: 1 - 50 $\frac{mm}{min}$ (Given)
  - Min Linear Speed (V_{min}) = 1 $\frac{mm}{min}$
  - Max Linear Speed (V_{max}): 50 $\frac{mm}{min}$
- $\varnothing_{min}$ = 0.107 in → Table Lead Screws: 3/8 - 8 STUB
  - $\varnothing_{min\ Table}$ = 0.265 in
  - Lead (P) = 0.125 $\frac{in}{rev}$

$$0.125 \frac{in}{rev}\left(\frac{25.4mm}{in}\right) = 3.175 \frac{mm}{rev}$$

Unknowns:

- W_{min} (RPM)
- W_{max} (RPM)

Solving For W_{min}:

$$W_{min} = \frac{V}{P} = \frac{(1 \frac{mm}{min})}{(3.175 \frac{mm}{rev})} = \boxed{0.315\ RPM}$$

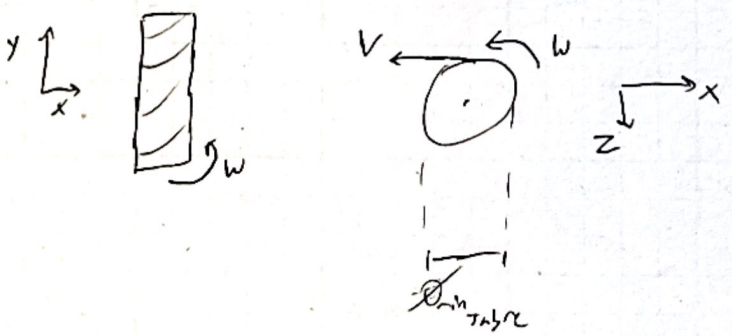
(eq 1)

Solving For W_{max}:

$$W_{max} = \frac{V}{P} = \frac{(50 \frac{mm}{min})}{(3.175 \frac{mm}{rev})} = \boxed{15.748\ RPM}$$

$$\boxed{\text{Range: } 0.315\ RPM - 15.748\ RPM}$$

FBD:



Determining Max Torque:

Knowns:
- Total Axial load (F_{Total}) = 2kN (Given)
- # of lead Screws (n) = 2 (Given)
- F_{lead} = 1kN
- $\phi_{min\ Table}$ = 0.265 in (From 3/8-8 STUB leadscrew)
- Wrange: 0.315 RPM - 15.748 RPM
- Coefficient of Friction (u) = 0.16 (Assumed)

Unknowns:
- T_{max}

Solving For T_{max}:

$$F = Q\left(\frac{P + 6.2832(u)(r)}{6.2832(r) - u(P)}\right) \times \frac{r}{R} \quad \text{(Friction Force)} \quad \text{(eq 1)}$$

$$F \times R = T = Q\left(\frac{P + 6.2832(u)(r)}{6.2832(r) - u(P)}\right)$$

Q = Load = 1000N = 224.809 lbf (per Screw)

P = Lead = 0.125 in/rev (Column C of lead Screw Table)

r = pitch radius: ϕ_{major} = 0.375 in (Column B)

$\phi_{min\ Table}$ = 0.265 in (Column G)

TPI = 8

$$d_{pitch} = \frac{\phi_{major} - \phi_{min\ Table}}{2} = \frac{0.375\text{in} - 0.265\text{in}}{2}$$

$$d_{pitch} = 0.160\text{ in}$$

$$r = \frac{\left[\frac{\phi_{major} - \phi_{min\ Table}}{2}\right]}{2} \quad \text{(eq 2)}$$

$$T_{max} = \frac{F_{Total}}{2}\left(\frac{P + 6.2832(u)\left(\frac{\left[\frac{\phi_{major} - \phi_{min\ Table}}{2}\right]}{2}\right)}{6.2832\left(\frac{\left[\frac{\phi_{major} - \phi_{min}}{2}\right]}{2}\right) - u(P)}\right)$$

$$T_{max} = 224.809\text{ lbf}\left(\frac{0.125\frac{\text{in}}{\text{rev}} + 6.2832(0.16)\left(\frac{\left[\frac{0.375\text{in} - 0.265\text{in}}{2}\right]}{2}\right)}{6.2832\left(\frac{\left[\frac{0.375\text{in} - 0.265\text{in}}{2}\right]}{2}\right) - (0.16)\left(0.125\frac{\text{in}}{\text{rev}}\right)}\right)$$

$$\boxed{T_{max} = 10.05\text{ in} \cdot \text{lbf}}$$

Scanned with CamScanner

Determining Max Torque Load

FBD:



Drive Torque Ratio (Column H)

$= 0.047 \frac{in\text{-}lb}{lb_f}$

T_{MAX} = individual max torque per screw

$T_{TOTAL\ max} = 2(T_{MAX})$

$T_{TOTAL\ max} = 2(10.05\ in\text{-}lb_f)$

$T_{TOTAL\ max} = 20.1\ in\text{-}lb_f$